

UNDERSPECIFIED PLAN

```

sequential([
  a(PickUpAction)(
    arm=LEFT,
    object_designator=
      a(DetectedMontessoriShape)(
        category=CUBE)
      .where(
        Colored(shape, CYAN),
        SupportedBy(shape, lid)),
    grasp_description=
      a(GraspDescription)(
        approach_direction=...,
        vertical_alignment=TOP,
        end_effector=LEFT_HAND)
  ),
  an(InsertionAction)(
    arm=LEFT,
    object_designator=shape,
    grasp_description=grasp,
    target=
      a(ShapeSortingHole)(
        shape_category=...)
        .from_(board.apertures)
  )
])

```

RESOLVED BY

PerceptionBackend

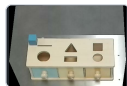
current view

cyan



on the lid

cube



spawn

cube_1

Simulation backend

imagined world

cube_1

overlap 3 mm $\leq$ 0.1 m

SupportedBy(cube_1, lid) $\rightarrow$ True

ProbabilisticBackend

0.58

P(grasp | success)

0.24

0.13

0.05

FRONT

LEFT

RIGHT

BACK

TOP

TOP

TOP

TOP

Ripple-down rules

shape.category == CUBE
 $\rightarrow$ hole.shape_category = CUBE

except

else

occupied(hole)
 $\rightarrow$ next free one

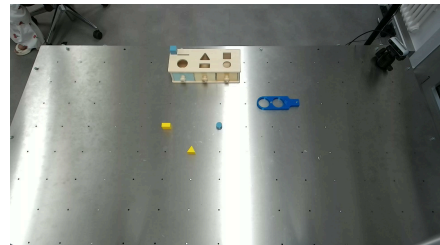
no rule fires
 $\rightarrow$ NoHoleFits

RESOLVED PLAN

```

sequential([
  PickUpAction(
    arm=LEFT,
    object_designator=
      cube_1 # CYAN, CUBE
      at (0.72, 0.13, 0.10) m,
      SupportedBy(cube_1, lid) ✓
    grasp_description=
      grasp_1 = GraspDescription(
        FRONT, TOP,
        LEFT_HAND)
  ),
  InsertionAction(
    arm=LEFT,
    object_designator=cube_1,
    grasp_description=grasp_1,
    target=
      ShapeSortingHole(
        shape_category=CUBE)
  )
])

```



Tracy

executes the resolved plan

